# Deep Residual Learning for Image Recognition and V isual Classification

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## ABSTRACT

*Deeper convolutional neural networks are more difficult to train. We present a residual learning framework to ease the training of networks that are substantially deeper than those used previously. We explicitly reformulate the layers as learning residual functions with reference to the layer inputs, instead of learning unreferenced functions. We provide comprehensive empirical evidence showing that these residual networks are easier to optimize, and can gain accuracy from considerably increased depth. On the ImageNet dataset we evaluate residual nets with a depth of up to 152 layers?8× deeper than VGG nets but still having lower complexity.*

## 1. INTRODUCTION

Deep convolutional neural networks have led to a series of breakthroughs for image classification. Driven by the significance of depth, recent leading results on the challenging ImageNet dataset all exploit 'very deep' models with a depth of 16 to 30. Many other visual recognition tasks have also greatly benefited from deep models. But when deeper networks are able to start converging, a degradation problem has been exposed: with network depth increasing, accuracy gets saturated and then degrades rapidly. Unexpectedly, such degradation is not caused by overfitting, and adding more layers to a suitably deep model leads to higher training error.

## 2. PROPOSED METHODOLOGY & SYSTEM ARCHITECTURE

Instead of hoping every few stacked layers directly fit a desired underlying mapping H(x), we explicitly let these layers fit a residual mapping F(x) := H(x) - x. The original mapping is recast into F(x) + x. We hypothesize that it is easier to optimize the residual mapping than to optimize the original, unreferenced mapping. The formulation of F(x) + x can be realized by feedforward neural networks with 'shortcut connections'. In our architecture, shortcut connections simply perform identity mapping, and their outputs are added to the outputs of the stacked layers. Identity shortcut connections introduce neither extra parameter nor computation complexity.

## 3. EXPERIMENTAL EVALUATION & RESULTS

We evaluate our methods on the ImageNet 2012 classification dataset that consists of 1000 classes. An ensemble of these residual nets achieves 3.57% top-5 error on the ImageNet test set. This result won first place in the ILSVRC 2015 classification competition. Our 152-layer ResNet achieves significantly higher accuracy than 34-layer plain networks while demonstrating smooth convergence throughout standard stochastic gradient descent training.

## 4. CONCLUSION AND FUTURE WORK

Residual learning effectively addresses the degradation problem that plagues extremely deep neural networks. By enabling depth up to 152 layers, ResNet establishes new state-of-the-art benchmarks in image recognition, object detection, and semantic segmentation.